

T1:

Forward

- Let $x_0 = 1.0, w_0 = 0.3, w_1 = -0.2, b_0 = 0.1, b_1 = -0.3$
- $u = x_0 w_0 + b_0 = 1(0.3) + 0.1 = 0.4$
- $x_1 = \text{ReLU}(u) = 0.4; 0.4 > 0$
- $y_1 = x_1 * w_1 + b_1 = 0.4 * (-0.2) + (-0.3) = -0.38$
- $z = \text{ReLU}(y_1 + x_0) = \text{ReLU}(-0.38 + 1) = \text{ReLU}(0.62) = 0.62$

Backward

- $\frac{dz}{d(y_1 + x_0)} = 1; \text{ReLU}(y_1 + x_0)$ มีค่าเป็น 1 ; ดังนั้น $\frac{dz}{dy_1} = 1, \frac{dz}{dx_0} = 1$
- จาก $y_1 = x_1 * w_1 + b_1 \rightarrow \frac{dy_1}{dw_1} = x_1 = 0.4, \frac{dy_1}{dx_1} = w_1 = -0.2, \frac{dy_1}{db_1} = 1$
- Chain rule: $\frac{dz}{dy_1} \frac{dy_1}{dw_1} = \frac{dz}{dw_1} = 0.4, \frac{dz}{dy_1} \frac{dy_1}{db_1} = \frac{dz}{db_1} = 1, \frac{dz}{dy_1} \frac{dy_1}{dx_1} = \frac{dz}{dx_1} = -0.2$
- จาก $x_1 = \text{ReLU}(x_0 w_0 + b_0) \rightarrow$ จาก $u = 0.4 > 0$ ดังนั้น $\text{ReLU}'(u) = 1 \rightarrow \frac{dx_1}{du} = 1$
- $\frac{dz}{dx_1} \frac{dx_1}{du} \frac{du}{dw_0} = \frac{dz}{dw_0} = -0.2 * 1 = -0.2, \frac{dz}{dx_1} \frac{dx_1}{du} \frac{du}{b_0} = \frac{dz}{b_0} = -0.2 * 1 = -0.2$

สรุป $\frac{dz}{dw_0} = -0.2, \frac{dz}{db_0} = -0.2, \frac{dz}{dw_1} = 0.4, \frac{dz}{db_1} = 1$

T2:

Input shape: (32, 30)

A: 1024 neuron -> output shape = (32, 1024)

B: 512 neuron -> output shape = (32, 512)

C: 1 neuron -> output shape = (32, 1)

T3:

Parameter = input * output + bias (1 bias/neuron)

Layer 1 : 30 feature * 1024 neuron + 1024 bias = 31744

Layer 2 : 1024 neuron_layer1 * 512 neuron_layer2 + 512 bias = 524800

Layer 3 : 512 neuron_layer2 * 1 neuron_layer3 + 1 bias = 513

รวม 3 layer = 31744 + 524800 + 513 = 557,057

T4:

$$\text{nn } P(y=j) = \frac{e^{h_j}}{\sum_k e^{h_k}} \rightarrow \text{nn } Z = \sum_k e^{h_k}, \frac{\partial Z}{\partial h_i} = e^{h_i} \cdot (k=i)$$

$$\therefore P(y=j) = \frac{e^{h_j}}{Z}$$

$$\frac{\partial P(y=j)}{\partial h_i}; \text{ case I } (j=i) \rightarrow \frac{\partial P(y=i)}{\partial h_i} = \frac{\partial}{\partial h_i} \left(\frac{e^{h_i}}{Z} \right) = \frac{Z e^{h_i} - e^{h_i} \cdot e^{h_i}}{Z^2} = \frac{e^{h_i}}{Z} - \left(\frac{e^{h_i}}{Z} \right)^2 = P_i(1-P_i)$$

$$\text{Case II } (j \neq i) \rightarrow \frac{\partial P(y \neq i)}{\partial h_i} = e^{h_i \neq i} \frac{\partial}{\partial h_i} \left(\frac{1}{Z} \right) = e^{h_i \neq i} \left(-\frac{1}{Z^2} \right) e^{h_i} = -\frac{e^{h_j}}{Z} \cdot \frac{e^{h_i}}{Z} = -P_j P_i$$

$$\text{nn } L = -\sum_j y_j \log P(y=j)$$

$$\frac{\partial L}{\partial h_i} = -\sum_j \frac{y_j}{P(y=j)} \cdot \frac{\partial P(y=j)}{\partial h_i}$$

$$\text{Case } j=i: \frac{\partial L}{\partial h_i} = \frac{-y_i}{P(y=i)} \cdot P_i(1-P_i) = -y_i + y_i P_i$$

$$\text{Case } j \neq i: \frac{\partial L}{\partial h_i} = \sum_{j \neq i} \frac{-y_j}{P(y \neq i)} \cdot (-P_j P_i) = \sum_{j \neq i} y_j P_i = P_i \sum_{j \neq i} y_j$$

$$\text{nn one-hot } \sum_j y_j = 1 \rightarrow \sum_{j \neq i} y_j = 1 - y_i \rightarrow \frac{\partial L}{\partial h_i} = P_i(1-y_i) = P_i - y_i P_i$$

$$\text{so case } j=i, j \neq i \rightarrow \frac{\partial L}{\partial h_i} = -y_i + y_i P_i + P_i - y_i P_i = P_i - y_i$$

$$\text{nn } \frac{\partial L}{\partial h_i} = P_i - y_i$$